# Nicolas Bailliet

+33 6 51 75 16 99 | nicolas.bailliet@etu.toulouse-inp.fr | Paris, France

baillietn.github.io | github.com/baillietn | linkedin.com/in/nbailliet | huggingface.co/baillietn

*Seeking a 6-month end-of-studies internship from mid-March 2027 · Open to relocation*

## Summary

Final-year Master's student in Computer Science at ENSEEIHT, with a background in mathematics and optimization. Built HilbertLM end to end on a single desktop GPU: tokenizer, pre-training on 20B tokens, fine-tuning, live chatbot. Looking to build, deploy and fine-tune LLM and agentic systems in production.

## Experience

**Research Engineer Intern — Optimal Control & Scientific Computing** Jun 2025 – Aug 2025

*IRIT / ENSEEIHT · Julia, JuMP* *Toulouse, France*

- Built an automated benchmark looping over 15 control problems: convergence checks for both the solver and JuMP (reference Julia modeling language), initialization comparison, and relative error in the $L^2$ norm on the state x and control u between the two solvers.
- Contributed unit tests & API documentation via GitHub (commits, Pull Requests, peer reviews), positive team feedback for surfacing critical bugs.

## Education

**ENSEEIHT (INP Toulouse)** Sep 2023 – Sep 2027 (expected)

*Diplôme d'Ingénieur (MSc equivalent), Computer Science & Engineering* *Toulouse, France*

ML algorithms, numerical methods, optimization, HPC, big data systems. Includes a required 6-month end-of-studies internship.

**Lycée Chaptal** Sep 2020 – Jul 2023

*Classes Préparatoires (CPGE), Mathematics & Physics* *Paris, France*

Intensive undergraduate program in mathematics and physics, preparing for the competitive entrance exams to French engineering schools.

## Projects

**HilbertLM (135M & 163M)**

baillietn.github.io/HilbertLM-Lab | huggingface.co/collections/baillietn/hilbertlm

*Python, PyTorch, Transformer Engine, Flash Attention 3, FP8, HuggingFace, BPE*

135M & 163M params | 20B tokens trained | ~70h · RTX 5080 | 79K tok/s throughput | 4 models on HF Hub

- Two decoder-only transformers built entirely from scratch. 30-layer architecture: GQA (9Q/3KV heads), RoPE, SwiGLU; custom BPE tokenizer (49K vocab).
- 4-stage curriculum pre-training (FineWeb-Edu, Python-Edu, FineMath, Cosmopedia-v2) + SFT on 100M conversational tokens (SmolTalk).
- Deployed as Base & Chat variants (BF16 + FP8) on HuggingFace Hub with a streaming chatbot interface.

**Soprasteria–ENSEEIHT · Water Quality IBD Prediction**

*Python, TensorFlow/Keras, scikit-learn, pandas*

Neural network predicting Biological Diatom Index for river ecosystem monitoring. 70+ environmental features from Naïades DB (2000–2025); BatchNorm & Dropout; anomaly detection & feature importance analysis.

## Skills

**Machine Learning & LLMs:** LLM architecture (GQA · RoPE · SwiGLU), Pre-training pipeline (data · curriculum · BPE), Fine-tuning (SFT), Mixed-precision training (BF16/FP8), Inference & deployment (HF)

**Frameworks & Tools:** Python, PyTorch, HuggingFace, Transformer Engine, Flash Attention, FP8 / BF16, JAX/Flax, NumPy, scikit-learn, Julia, SQL, Git, Linux, HPC

**Languages:** French (native), English (professional)